Ashutosh Pal Singh

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PROFILE

Cyber security graduate with an MSc in Cyber Security Engineering from the University of Warwick and the Microsoft SC-200 certification. Builds and validates detections mapped to MITRE ATT&CK and triages phishing, endpoint and Active Directory activity in a private Elastic Stack, Splunk and Sysmon lab. Maintained production security monitoring and ran vulnerability assessments across an IoT device estate, automating the cycle in Python.

EXPERIENCE

Cyber Security Analyst **Jun 2024 to Dec 2024**
Kyoka Embedded Systems Pvt Ltd (smart home IoT device manufacturer), seven month internship *Rajkot, India*

- Maintained day to day operation of the security monitoring infrastructure at 99.9% uptime over the seven months, and supported the rollout of mutual TLS device authentication with hardware backed key storage.
- Automated scan scheduling, result parsing and reporting in Python, removing manual steps from the recurring assessment cycle.
- Ran the end to end assessment cycle with Nessus and Qualys across a production IoT estate of 1,000+ smart home devices: scanning, CVSS prioritisation, remediation tracking and verification on device.
- Reduced open vulnerabilities on the assessed estate by 45% between baseline and final scans over six months, delivering vulnerability reports and remediation guidance to engineering and management.
- Identified five bootloader level flaws through hardware and firmware analysis and drove the fixes with the embedded engineering team, verifying each fix on device.

Cyber Security Research Analyst **Jan 2024 to May 2024**
NASSCOM (India's national technology industry body), policy research internship *Noida, India*

- Drafted, as joint author, a policy document on deepfake and synthetic media governance, and contributed research to briefings on cyber incident reporting obligations.
- Tested C2PA, Google SynthID and Meta Stable Signature hands on within a survey of 250+ tools and standards, mapping the findings to the UK Online Safety Act and the EU AI Act.

Network Security Engineering, Cisco AICTE Virtual Internship Programme (simulation based) **Apr 2023 to Jul 2023**
Cisco Networking Academy *Online*

- Designed a 50+ node network across multiple sites in Cisco Packet Tracer with VLAN segmentation, firewall rules and ACLs, then audited it and ranked 15+ weaknesses by risk.

Penetration Tester (six week training programme) **Jul 2022 to Aug 2022**
Verzeo (online training provider) *Online*

- Tested 10+ lab Windows targets with Nmap and Metasploit on Kali Linux, reported findings against the OWASP Top 10 and ran phishing simulations with the Social Engineer Toolkit.

EDUCATION

MSc Cyber Security Engineering (elective: AI for Cyber Security) **Sep 2025 to Sep 2026**
University of Warwick, WMG (NCSC certified programme) *Coventry, UK*
Dissertation submitted Sep 2026; result pending. Modules: Proactive Cyber Defence, Applied Cryptography, Managing Cyber Risk Audit and Compliance, Ethical Hacking, Digital Forensics, Machine Learning for Cyber Security.

BTech Computer Science Engineering **Aug 2020 to Jun 2024**
Symbiosis Institute of Technology *Pune, India*

PROJECTS

Detection Engineering Home Lab **Jan 2025 to present**
Private build; walkthrough available at interview *Elastic Stack, Splunk, Sysmon, Sigma*

- Built an Elastic Stack and Sysmon telemetry pipeline on hardened Debian hosts to collect and analyse Windows endpoint and Active Directory logs.
- Wrote Sigma and Splunk rules mapped to MITRE ATT&CK techniques and validated each against lab telemetry so it fires on attack activity and stays quiet on normal use.
- Emulated adversary techniques on a lab Active Directory (Kerberoasting, BloodHound, Impacket) to measure detection coverage, and triaged the resulting PCAPs in Wireshark.
- Triaged phishing samples through SPF, DKIM and DMARC header analysis, enriched indicators with VirusTotal, Shodan, AbuseIPDB and YARA, and practised Microsoft Sentinel and KQL during SC-200 preparation.

CERTIFICATIONS AND PUBLICATION

- Microsoft Certified: Security Operations Analyst Associate (SC-200), 2026.
- Google Cybersecurity Professional Certificate, Apr 2024. EC-Council Digital Forensics Essentials (DFE).
- Singh, A.P. (2024) "Safeguarding Authenticity in the Digital Realm: Content Provenance and Watermarking to Combat Deepfakes", International Journal for Multidisciplinary Research, 6(3).

TECHNICAL SKILLS

Detection and response: Splunk, Elastic Stack, Sysmon, Sigma, MITRE ATT&CK, Microsoft Sentinel, KQL, Defender XDR
Investigation and forensics: Wireshark, Volatility, Autopsy, FTK Imager, VirusTotal, Shodan, AbuseIPDB, YARA
Scripting and systems: Python, SQL, Bash, Linux, Windows, Git, Nessus, Qualys, CVSS, Nmap